Quantum Kerr OAM

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An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. THEORY

Consider a bosonic quantum field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$\left[\mathcal{E}(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)\right] = \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2). \tag{1}$$

In the above equation and in what follows, $\mathbf{r} = (x, y)$ is the transverse coordinate. We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z \mathcal{E}(\mathbf{r}, z) + \nabla^2 \mathcal{E}(\mathbf{r}, z) = -g\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}^2(\mathbf{r}, z). \tag{2}$$

Let us decompose $\mathcal{E} = \mathcal{E}_0 + \delta\mathcal{E}$ where $\mathcal{E}_0$ is the solution of

$$2ik\partial_z \mathcal{E}_0(\mathbf{r}, z) + \nabla^2 \mathcal{E}_0(\mathbf{r}, z) = 0, \quad \mathcal{E}_0(\mathbf{r}, 0) = \mathcal{E}(\mathbf{r}, 0). \tag{3}$$

Let u be a solution of the paraxial wave equation $2ik\partial_z u(\mathbf{r}, z) + \nabla^2 u(\mathbf{r}, z) = 0$. Then, the anihilation operator corresponding to mode u is given by

$$a_u(z) = \int u^*(\mathbf{r}, z)\mathcal{E}(\mathbf{r}, z)d\mathbf{r} = a_{u,0} + \delta a_u(z) \tag{4}$$

where

$$a_{u,0} = \int u^*(\mathbf{r}, z)\mathcal{E}_0(\mathbf{r}, z)d\mathbf{r} \tag{5}$$

is independent of z (take a z derivative, use de equations of motion and integrate by parts) and

$$\delta a_u(z) = \int u^*(\mathbf{r}, z)\delta\mathcal{E}(\mathbf{r}, z)d\mathbf{r} \tag{6}$$

Our initial state will be a coherent one:

$$|\alpha, u\rangle = D(\alpha, u)\,|0\rangle = \exp\left(\alpha a_{u,0}^\dagger - \alpha^* a_{u,0}\right)|0\rangle\,. \tag{7}$$

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This state is an eigenvector of the free field:

$$\mathcal{E}_0(\mathbf{r},z)\left|\alpha,u\right\rangle = \alpha u(\mathbf{r},z)\left|\alpha,u\right\rangle \tag{8}$$

The variation $\delta\mathcal{E}$ satisfies

$$2ik\partial_z\delta\mathcal{E}(\mathbf{r},z) + \nabla^2\delta\mathcal{E}(\mathbf{r},z) = -g\mathcal{E}^\dagger(\mathbf{r},z)\mathcal{E}^2(\mathbf{r},z). \tag{9}$$

Evaluating this at $z=0$ gives us

$$2ik\partial_z\delta\mathcal{E}(\mathbf{r},0) = -g\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{10}$$

from which we obtain the Taylor expansion

$$\delta\mathcal{E}(\mathbf{r},z) \approx \frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \approx \frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r},z)\mathcal{E}_0^2(\mathbf{r},z) \tag{11}$$

For the complete field we then have

$$\mathcal{E}(\mathbf{r},z) \approx \mathcal{E}_0(\mathbf{r},z) + \frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r},z)\mathcal{E}_0^2(\mathbf{r},z) \tag{12}$$

This expression will be the basis of our further developments.

A. Expectation value of field

Taking expectation values, we arrive at

$$\langle\mathcal{E}(\mathbf{r},z)\rangle \approx \alpha u(\mathbf{r},z) + \frac{igz|\alpha|^2\alpha}{2k}|u(\mathbf{r},z)|^2 u(\mathbf{r},z). \tag{13}$$

The expectation value of the annihilation operator a_v corresponding to a spatial mode v is then

$$\langle a_v\rangle = \alpha\left\langle v|u\right\rangle + \frac{igz|\alpha|^2\alpha}{2k}\left\langle v,u|u,u\right\rangle \tag{14}$$

In the above expression we have introduced the first spatial overlap

$$\langle v|u\rangle = \int v^*(\mathbf{r},z)u(\mathbf{r},z)d\mathbf{r} \tag{15}$$

and the second spatial overlap

$$\langle v_1,v_2|u_1,u_2\rangle = \int v_1^*(\mathbf{r},z)v_2^*(\mathbf{r},z)u_1(\mathbf{r},z)u_2(\mathbf{r},z)d\mathbf{r}. \tag{16}$$

When $u=u_{0l}$ is a Laguerre-Gaussian mode with radial order $p=0$ and topological charge l, we have that

$$|u_{0l}(\mathbf{r},z)|u_{0l}^2(\mathbf{r},z) = \sum_{q=0}^{|l|} C_{ql}\overline{u}_{ql}(\mathbf{r},z) \tag{17}$$

where $\overline{u}_{ql}$ are Laguerre-Gaussian modes with a reduced waist. Therefore, one may write

$$\langle a_v\rangle = \alpha\left\langle v|u_{0,l}\right\rangle + \frac{igz|\alpha|^2\alpha}{2k}\sum_{q=0}^{|l|} C_{ql}\left\langle v|\overline{u}_{ql}\right\rangle \tag{18}$$

In particular, when $v=\overline{u}_{ql}$, we the have that

$$\left\langle a_{\overline{u}_{ql}}\right\rangle = \begin{cases} \alpha\left\langle\overline{u}_{ql}|u_{0,l}\right\rangle + \dfrac{igz|\alpha|^2\alpha C_{ql}}{2k} & q\le|l| \\ \alpha\left\langle\overline{u}_{ql}|u_{0,l}\right\rangle & q>|l| \end{cases}. \tag{19}$$

B. Expectation value of number

The product of the field at two points is

$$\mathcal{E}^\dagger(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) + \mathcal{E}_0^\dagger(\mathbf{r}_1,z)\delta\mathcal{E}(\mathbf{r}_2,z) + \delta\mathcal{E}^\dagger(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) \tag{20}$$

where we have only kept terms linear in $\delta\mathcal{E}$.

Close to $z=0$, we have that

$$\mathcal{E}^\dagger(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) + \frac{igz}{2k}\left[\mathcal{E}_0^\dagger(\mathbf{r}_1,z)\mathcal{E}_0^\dagger(\mathbf{r}_2,z)\mathcal{E}_0^2(\mathbf{r}_2,z) - \mathcal{E}_0^\dagger(\mathbf{r}_1,z)^2\mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z)\right] \tag{21}$$

Evaluating this at our initial state gives us

$$\left\langle\mathcal{E}^\dagger(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\right\rangle \approx |\alpha|^2 u^*(\mathbf{r}_1,z)u(\mathbf{r}_2,z)\left\{1 + \frac{igz|\alpha|^2}{2k}\left[|u(\mathbf{r}_2,z)|^2 - |u(\mathbf{r}_1,z)|^2\right]\right\} \tag{22}$$

The number operator of mode v can be written in terms of this:

$$n_v = a_v^\dagger a_v = \int v(\mathbf{r}_1)v^*(\mathbf{r}_2)\mathcal{E}^\dagger(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)d\mathbf{r}_1 d\mathbf{r}_2 \tag{23}$$

The expectation value is then

$$\langle n_v\rangle \approx |\alpha\langle v|u\rangle|^2 + \frac{gz|\alpha|^4}{k}\operatorname{Im}\langle v|u\rangle\langle u,u|u,v\rangle \approx |\langle a_v\rangle|^2. \tag{24}$$

When both u and v are arbitrary Laguerre-Gaussian modes (any p,l,w), we have that $\operatorname{Im}\langle v|u\rangle\langle u,u|u,v\rangle = 0$, so there is no first order correction to the occupation number.

C. Quadratures

Let us define the quadrature associated to the spatial mode v as

$$\begin{aligned} X_v &= \frac{1}{\sqrt{2}}\int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r}) + v(\mathbf{r})\mathcal{E}^\dagger(\mathbf{r})d\mathbf{r} \\ &= \frac{a_v + a_v^\dagger}{\sqrt{2}} \end{aligned} \tag{25}$$

We then have the commutation relation

$$[X_{v_1}, X_{v_2}] = i\operatorname{Im}\langle v_1|v_2\rangle, \tag{26}$$

which leads to the uncertainty principle

$$\Delta X_{v_1}^2 \Delta X_{v_2}^2 \geq \frac{|\operatorname{Im}\langle v_1|v_2\rangle|^2}{4}. \tag{27}$$

One could also introduce the complementary quadrature $P(v) = X(-iv)$ so that

$$[X_{v_1}, P_{v_2}] = i\operatorname{Re}\langle v_1|v_2\rangle \tag{28}$$

and

$$\Delta X_{v_1}^2 \Delta P_{v_2}^2 \geq \frac{|\operatorname{Re}\langle v_1|v_2\rangle|^2}{4}. \tag{29}$$

We are interested in observables of the form

$$C(v_1,v_2) = \langle X_{v_1}X_{v_2}\rangle - \langle X_{v_1}\rangle\langle X_{v_2}\rangle = \frac{\langle v_1|v_2\rangle}{2} + \operatorname{Re}\left[\langle a_{v_1}a_{v_2}\rangle - \langle a_{v_1}\rangle\langle a_{v_2}\rangle + \left\langle a_{v_2}^\dagger a_{v_1}\right\rangle - \left\langle a_{v_2}^\dagger\right\rangle\langle a_{v_1}\rangle\right]. \tag{30}$$

For the Kerr system, we have

$$\begin{aligned}\left\langle a_{v_2}^\dagger a_{v_1}\right\rangle &= \int v_2(\mathbf{r}_2)v_1^*(\mathbf{r}_1)\left\langle \mathcal{E}^\dagger(\mathbf{r}_2,z)\mathcal{E}(\mathbf{r}_1,z)\right\rangle d\mathbf{r}_1 d\mathbf{r}_2 \\ &\approx |\alpha|^2 \langle v_1|u\rangle \langle u|v_2\rangle + \frac{gz|\alpha|^4}{k}\operatorname{Im}\langle v_1|u\rangle \langle u,u|u,v_2\rangle \\ &\approx \left\langle a_{v_2}^\dagger\right\rangle \langle a_{v_1}\rangle\end{aligned} \tag{31}$$

So the last two terms in (30) do not contribute. On the other hand, we have that

$$\begin{aligned}\mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z) &\approx \mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) + \mathcal{E}_0(\mathbf{r}_1,z)\delta\mathcal{E}(\mathbf{r}_2,z) + \delta\mathcal{E}(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) \\ &\approx \mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) + \frac{igz}{2k}\left[\mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0^\dagger(\mathbf{r}_2,z)\mathcal{E}_0^2(\mathbf{r}_2,z) + \mathcal{E}_0^\dagger(\mathbf{r}_1,z)\mathcal{E}_0^2(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z)\right] \\ &= \mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) + \frac{igz}{2k}\left[\mathcal{E}_0^\dagger(\mathbf{r}_2,z)\mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0^2(\mathbf{r}_2,z) + \mathcal{E}_0^\dagger(\mathbf{r}_1,z)\mathcal{E}_0^2(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) + \delta(\mathbf{r}_1-\mathbf{r}_2)\mathcal{E}_0^2(\mathbf{r}_2,z)\right]\end{aligned} \tag{32}$$

The appearance of the delta function here is important, because now

$$\langle \mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle \approx \alpha^2 u(\mathbf{r}_1,z)u(\mathbf{r}_2,z)\left[1 + \frac{igz|\alpha|^2}{2k}\left[|u(\mathbf{r}_2,z)|^2 + |u(\mathbf{r}_1,z)|^2\right]\right] + \frac{igz\alpha^2}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)u^2(\mathbf{r}_2,z) \tag{33}$$

$$\begin{aligned}\langle a_{v_1}a_{v_2}\rangle &= \int v_1^*(\mathbf{r}_1)v_2^*(\mathbf{r}_2)\langle \mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle d\mathbf{r}_1 d\mathbf{r}_2 \\ &\approx \alpha^2\langle v_1|u\rangle\langle v_2|v_1\rangle + \frac{igz|\alpha|^2\alpha^2}{2k}\left(\langle v_1|u\rangle\langle v_2,u|u,u\rangle + \langle v_2|u\rangle\langle v_1,u|u,u\rangle\right) + \frac{igz\alpha^2}{2k}\langle v_1,v_2|u,u\rangle \\ &\approx \langle a_{v_2}\rangle\langle a_{v_1}\rangle + \frac{igz\alpha^2}{2k}\langle v_1,v_2|u,u\rangle\end{aligned} \tag{34}$$

We then finally get that

$$C(v_1,v_2) \approx \frac{\langle v_1|v_2\rangle}{2} - \frac{gz\alpha^2}{2k}\operatorname{Im}\langle v_1,v_2|u,u\rangle, \tag{35}$$

where we have chosen to measure phases with respect to the input state ($\operatorname{Im}\alpha = 0$).

Let us the calculate the matrix $C_{jk} = \langle Z_j Z_k\rangle - \langle Z_j\rangle\langle Z_k\rangle$ with $Z = X_{v_1}, P_{v_1}, X_{v_2}, P_{v_2}$, in this order. We have that

$$C = \begin{pmatrix} C(v_1,v_1) & C(v_1,-iv_1) & C(v_1,v_2) & C(v_1,-iv_2) \\ C(-iv_1,v_1) & C(-iv_1,-iv_1) & C(-iv_1,v_2) & C(-iv_1,-iv_2) \\ C(v_2,v_1) & C(v_2,-iv_1) & C(v_2,v_2) & C(v_2,-iv_2) \\ C(-iv_2,v_1) & C(-iv_2,-iv_1) & C(-iv_2,v_2) & C(-iv_2,-iv_2)\end{pmatrix} \tag{36}$$

We now assume that $\langle v_1|v_2\rangle = 0$. We then have $C = C_0 + \delta C$ where

$$C_0 = \frac{1}{2}\begin{pmatrix} 1 & -i & 0 & 0 \\ i & 1 & 0 & 0 \\ 0 & 0 & 1 & -i \\ 0 & 0 & i & 1\end{pmatrix}; \quad \delta C = \frac{gz\alpha^2}{2k}\begin{pmatrix} -\operatorname{Im}\langle v_1,v_1|u,u\rangle & -\operatorname{Re}\langle v_1,v_1|u,u\rangle & -\operatorname{Im}\langle v_1,v_2|u,u\rangle & -\operatorname{Re}\langle v_1,v_2|u,u\rangle \\ -\operatorname{Re}\langle v_1,v_1|u,u\rangle & \operatorname{Im}\langle v_1,v_1|u,u\rangle & -\operatorname{Re}\langle v_1,v_2|u,u\rangle & \operatorname{Im}\langle v_1,v_2|u,u\rangle \\ -\operatorname{Im}\langle v_1,v_2|u,u\rangle & -\operatorname{Re}\langle v_1,v_2|u,u\rangle & -\operatorname{Im}\langle v_2,v_2|u,u\rangle & -\operatorname{Re}\langle v_1,v_1|u,u\rangle \\ -\operatorname{Re}\langle v_1,v_2|u,u\rangle & \operatorname{Im}\langle v_1,v_2|u,u\rangle & -\operatorname{Re}\langle v_1,v_1|u,u\rangle & \operatorname{Im}\langle v_2,v_2|u,u\rangle\end{pmatrix} \tag{37}$$

At this point, it becomes clear that the overlaps of the form $\langle v_1,v_2|u,u\rangle$ are the ones that dominate this sort of process. On the other hand, the generation of radial orders comes from the properties of the different overlap $\langle v,u|u,u\rangle$, which appears in (14). It is therefore hard to see any special implication of the selection rules here.

Let us restrict our attention to vortices: $u(\mathbf{r}) = u_r(r)e^{il\phi}; v_j(\mathbf{r}) = v_{j,r}(r)e^{il_j\phi+\pi/4}$ where the radial functions are real valued. We further assume that $l_1 + l_2 = 2l$. Then, all the real terms in δC disappear and we get

$$\delta C = \frac{gz\alpha^2}{2k}\begin{pmatrix} -R_{11}\delta_{l_1,l_2} & 0 & -R_{12} & 0 \\ -0 & R_{11}\delta_{l_1,l_2} & 0 & R_{12} \\ -R_{12} & 0 & -R_{22}\delta_{l_1,l_2} & 0 \\ 0 & R_{12} & 0 & R_{22}\delta_{l_1,l_2}\end{pmatrix}, \quad R_{jk} = 2\pi\int_0^\infty v_{r,j}(r)v_{r,k}(r)u_r^2(r)dr \tag{38}$$

When $l_1 \neq l_2$, the variance in each mode does not change, but we see correlations between the quadratures of different modes induced by R_{12}. When $l_1 = l_2$, we also see maximum squeezing along X_{v_j} and antisqueezing along P_{v_j}.

The EPR correlations are given by

$$\left\langle \Delta(X_{v_1} \pm X_{v_2})^2 \right\rangle + \left\langle \Delta(P_{v_1} \mp P_{v_2})^2 \right\rangle = 2 \mp \frac{gz\alpha^2}{k} R_{12} \tag{39}$$

Then, the state is entangled according to the Duan criterion [1].

II. NUMERICAL CONSIDERATIONS

Note that, according to (1), the field $\mathcal{E}(\mathbf{r})$ has dimensions of $1/\sqrt{\text{area}}$. Therefore, by dimensional analysis on eq. (2), we have that g is dimensionless.

In order to put the equation in a more convenient form, we introduce the dimensionless coordinate $\tilde{\mathbf{r}} = \mathbf{r}/w_0$ where w_0 is the waist of the input beam. We also introduce the dimensionless propagation distance $\tilde{z} = z/z_R$ where $z_R = kw_0^2/2$ is the Rayleigh range. Finally, we introduce the rescaled field $\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = \sqrt{\Delta A}\mathcal{E}(\mathbf{r}, z)$, where ΔA is the area of a pixel in the numerical grid. In terms of these variables, the commutation relations read

$$\left[\tilde{\mathcal{E}}(\tilde{\mathbf{r}}), \tilde{\mathcal{E}}(\tilde{\mathbf{r}})^\dagger\right] \approx \delta_{\tilde{\mathbf{r}},\tilde{\mathbf{r}}'}, \tag{40}$$

while the equation of motion reads

$$i\partial_{\tilde{z}}\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = -\frac{1}{4}\tilde{\nabla}^2\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) - \frac{g}{4\Delta\tilde{A}}\tilde{\mathcal{E}}^\dagger(\tilde{\mathbf{r}}, \tilde{z})\tilde{\mathcal{E}}^2(\tilde{\mathbf{r}}, \tilde{z}) \tag{41}$$

where $\Delta\tilde{A} = \Delta A/w_0^2$ is the area of a pixel in the rescaled coordinates.

In the positive P representation, the field $\tilde{\mathcal{E}}$ is represented by two independent complex stochastic fields ψ_1 and ψ_2 satisfying the stochastic equations (compare with [2])

$$\begin{aligned} i\partial_{\tilde{z}}\psi_1(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_1(\tilde{\mathbf{r}}, \tilde{z}) - \frac{g}{4\Delta\tilde{A}}\psi_2(\tilde{\mathbf{r}}, \tilde{z})\psi_1^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{\frac{ig}{4\Delta\tilde{A}}}\psi_1(\tilde{\mathbf{r}}, \tilde{z})\xi_1(\tilde{\mathbf{r}}, \tilde{z}) \\ -i\partial_{\tilde{z}}\psi_2(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_2(\tilde{\mathbf{r}}, \tilde{z}) - \frac{g}{4\Delta\tilde{A}}\psi_1(\tilde{\mathbf{r}}, \tilde{z})\psi_2^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{\frac{ig}{4\Delta\tilde{A}}}\psi_2(\tilde{\mathbf{r}}, \tilde{z})\xi_2(\tilde{\mathbf{r}}, \tilde{z}) \end{aligned} \tag{42}$$

where ξ_1 and ξ_2 are independent real Gaussian white noises with zero mean and correlations

$$\langle \xi_j(\tilde{\mathbf{r}}, \tilde{z})\xi_k(\tilde{\mathbf{r}}', \tilde{z}')\rangle = 0, \quad \langle \xi_j(\tilde{\mathbf{r}}, \tilde{z})\xi_k(\tilde{\mathbf{r}}', \tilde{z}')\rangle = \delta_{jk}\delta_{\tilde{\mathbf{r}},\tilde{\mathbf{r}}'}\delta_{\tilde{z},\tilde{z}'} \tag{43}$$

For a coherent initial state in the mode u, we have that $\psi_1(\tilde{\mathbf{r}}, 0) = \alpha u(\tilde{\mathbf{r}}, 0)$ and $\psi_2(\tilde{\mathbf{r}}, 0) = \alpha^* u^*(\tilde{\mathbf{r}}, 0)$.

[1] L.-M. Duan, G. Giedke, J. I. Cirac, and P. Zoller, Inseparability criterion for continuous variable systems, Physical review letters **84**, 2722 (2000).

[2] P. Deuar and P. D. Drummond, Correlations in a bec collision: First-principles quantum dynamics with 150 000 atoms, Phys. Rev. Lett. **98**, 120402 (2007).